Human Body Systems: Digestion & Control

Quick reference sheet · Biology · keep handy for revision

1. The digestive system

Digestion breaks down large, insoluble food molecules into small, soluble molecules that can be absorbed into the blood.

Organ	What it does
Mouth	Chews food; saliva contains amylase, which starts digesting starch
Oesophagus	Muscular tube that pushes food down to the stomach
Stomach	Churns food; produces acid (kills bacteria, right pH for protease) and protease
Small intestine	Digestion completed here; digested food is absorbed into the blood
Large intestine	Absorbs water; forms faeces
Pancreas	Produces amylase, protease & lipase, released into the small intestine
Liver	Produces bile

2. Enzymes & bile

Enzymes are **biological catalysts** — proteins that speed up reactions without being used up. Each has an active site shaped to fit one substrate (lock and key).

Enzyme	Breaks down	Into
Amylase	Starch	Sugars
Protease	Protein	Amino acids
Lipase	Lipids (fats)	Fatty acids + glycerol

Bile (made in the liver) is not an enzyme. It **neutralises** stomach acid, giving enzymes in the small intestine the right (alkaline) pH, and **emulsifies** fats into tiny droplets, increasing the surface area for lipase.

3. The nervous system & reflexes

A **reflex** is a fast, automatic response that does not involve conscious thought. Pathway:

stimulus → **receptor** → **sensory neuron** → **relay neuron (spinal cord)** → **motor neuron** → **effector** → **response**

A **synapse** is the gap between two neurons; a chemical is released to carry the signal across it. Reflexes are fast because the signal is processed in the spinal cord, not the brain.

4. The endocrine system

Hormones are chemical messengers released directly into the blood by glands, producing an effect at a target organ. Slower to start than nerves, but longer-lasting.

Gland	Hormone	Effect
Pituitary gland	Several	"Master gland" — controls other glands
Pancreas	Insulin	Reduces blood glucose concentration
Adrenal glands	Adrenaline	"Fight or flight" response
Thyroid gland	Thyroxine	Regulates metabolic rate

Nervous vs hormonal: nervous signals are electrical, travel along neurons — fast, short-lived. Hormonal signals are chemical, travel in the blood — slower, longer-lasting.